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Full Legal Name: Mohammad Marsad Salfi

Institution: St. Joseph Higher Secondary School

Category: Junior

Class/Grade/Year: Class 09

Country: Bangladesh

Submission Date: September 20, 2026

Registered Email Address: sj20-3bv-106@sjs.edu.bd

Research Topic: Harnessing Agricultural Residues through Co-Gasification: A Techno-Economic Assessment for Energy Security in the Hilly Tracts of Bangladesh

Harnessing Agricultural Residues through Co-Gasification: A Techno-Economic Assessment for Energy Security in the Hilly Tracts of Bangladesh

Introduction

As a developing nation, Bangladesh has seen demand skyrocket. Fueled by its growing population, demand shows no signs of slowing down. As a fossil fuel-dependent country, Bangladesh can use agricultural waste as a substitute [1]. Together, the three waste streams (rice husk, jute stick, and banana peel) generate 12.3 to 14.6 million metric tons annually in Bangladesh. The total agricultural crop biomass nationwide reaches 65 to 70 million metric tons per year [2]. Due to inadequate management practices, these biomass residues remain largely unutilized across most sectors. This waste can be used to produce biogas using a downdraft gasifier. By using this downdraft gasifier, we can produce low-tar, high-quality syngas [3]. This can be used to produce clean energy. This study shows how underutilized agricultural residues can be converted into clean bioenergy, offering a sustainable dual solution for waste management and energy security.

Existing literature

While biomass gasification is a well-studied topic, most research focuses on industrial-scale equipment and advanced chemicals, creating a major knowledge gap for people living in rural or hilly regions. Most biological pretreatments require purified enzymes that depend on sterile labs and cold storage facilities. Because these resources are rarely available in rural areas, a major gap remains in practical methods for converting local organic waste into biogas.

In Bangladesh, this challenge is evident in the management of rice husks and jute waste. Agricultural activity generates between 9 and 12 million metric tons of rice husks and up to 3 million metric tons of jute sticks annually [4]. The rural area lacks access to urban laboratory equipment or specialized enzymatic treatments; rural populations overwhelmingly burn these residues for household cooking, which causes uncontrollable environmental pollution and public health risks. Addressing this issue requires chemical-free, accessible processing methods specifically designed for the structural characteristics of Bangladeshi agricultural waste.

Biomass	Electricity yield	Basis
Rice husk	**about 538 kWh/ton**	Gasification
Sugarcane bagasse	**about 230–550 kWh/ton**	Cogeneration; technology dependent
Jute stalk	**about 300–500 kWh/ton**	Estimated from reported energy content and biomass-to-power conversion

These are the common biomass sources found around Bangladesh. Adapted from [5],[6],[7],[8].

Methodology

Materials

- **Dry Biomass Fuel** (rice husk, jute stick, or bagasse)
- **Gasifier Reactor** (refractory-lined chamber)
- **Air Blower** (supplies reaction air)
- **Gas Filter & Cooler** (removes ash, tar, and moisture)
- **Dual-Fuel Generator Set** (modified engine + alternator)

Setup

Taking inspiration from other small gasifiers (Belonio rice husk gasifier stove), we need to gather locally available materials like steel drums , pipes, and grates, and focus on trying to depend on welding or other heavy duty machinery which can make this disaster-prone or dependent.

Gathering materials/making the materials ready

Biomass feedstocks can adapt to regional supply. In Bangladesh, locally sourced rice husks, banana peels, bamboo stalks, and jute are collected from households, sun-dried, and manually chopped to manageable sizes

Reactor construction and setup

1. Gasifier Construction: A downdraft gasifier will be fabricated from locally available materials, consisting of an outer metal drum, an inner combustion–

reduction chamber, side air inlets positioned above the grate (ensuring downward gas flow), and an ash collection grate at the base.

2. Gas Cleaning: The raw producer gas will be passed through a low-cost cleaning unit — either a water scrubber or a charcoal/cloth filter — to remove tar and particulates before entering the engine, preventing fouling and clogging.
3. Energy Output: The cleaned gas will be routed to a small (1–5 kVA) dual-fuel generator for electricity, with diesel substitution expected at 70–85%, or directed to a modified household burner for cooking applications.

Experimental testing

- Blend: Trying different types of blends of different organic matter. The different moisture levels of the blends can behave differently within the experiment.
- Reactor Temperature: Monitored using a K-type thermocouple and digital reader.
- Combustion Performance: Evaluated using standard Water Boiling Test (WBT) protocols to measure flame quality and time required to boil 1 L of water.
- Char & Ash Yield: Measured by collecting and weighing remaining solid byproducts after each run.
- Generator Power Output: Recorded voltage, current, and total runtime on producer gas using a standard multimeter and clamp meter.
- Gas Analysis: Sent to a partner university lab for single-sample gas chromatography (CO, H, CH4), or proxied using flame heat and engine stability (and cited in the paper's limitations).

Safety Consideration

Because producer gas contains carbon monoxide, all tests were conducted outdoors with enforced leak checks and ignition-delay procedures, and indoor deployment guidelines mandate adequate ventilation. Solid residues are managed as co-products: biochar is recovered for use as a soil amendment or briquette feedstock, while ash is collected separately owing to the high silica content of rice husk.

Data collection and analysis

- Time until first gas is observed
- How the different biomass reacts together
- How much is generated from a specific batch of biomass

Research Problem

1. How effective is the low-cost drum gasifier in terms of: a. Temperature regulation during gasification; b. Tar removal efficiency using a simple water scrubber or charcoal filter; c. Syngas production rate and continuous flame stability; d. Char and ash byproduct yield
2. How effective is the energy output of the system in terms of: a. Time required to boil 1 L of water using a standard household burner (Water Boiling Test); b. Electrical power generation (voltage, current, and runtime) using small dual-fuel generators; c. Thermal efficiency compared to traditional earthen cookstoves
3. Is there a significant difference in the performance of the gasifier depending on the feedstock used in terms of: a. Syngas quality and flame temperature (Rice Husk vs. Jute Stick vs. Mixed Biomass); b. Operating runtime per fuel batch; c. Solid residue (char and ash) accumulation rate
4. What is the rate of acceptability of the decentralized biomass gasification system among rural households in terms of: a. Functionality b. Durability c. Accessibility

Expected Outcome

A small-scale biomass gasifier can use agricultural residues such as rice husks and jute sticks as fuel. These materials can generate around 3.0 to 8.0 kW of thermal power while consuming approximately 1.5 to 3.0 kg of biomass per hour. The gasifier produces syngas with a lower heating value of about 4.0 to 5.5 MJ/Nm3 [9], [10]. When the gasifier is connected to a small dual-fuel engine-generator, the system can achieve an overall electrical efficiency of approximately 12% to 18%. Under these conditions, it can produce around 1.0 to 3.0 kW of electrical power, with a specific fuel consumption of about 1.2 to 1.8 kg/kWh [11], [12].

Conclusion

A drum gasifier will power lights/phone charging/a small generator — not a whole village. This is only meant for places where electricity is non-existent or during natural disasters when conventional electricity transport is damaged. The expected outcome of this is supplementary energy using Bangladeshi organic waste. Ultimately, this study aims to demonstrate that clean electricity can be generated from agricultural waste using nothing more sophisticated than a steel drum, a filter, and locally available skills.

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